

Part 1:

1. A screenshot that shows you can successfully configure and compile ChampSim .

```
root@aaefef21d69a:/champsim# ./config.sh lab1_config.json
root@aaefef21d69a:/champsim# make -j $(nproc)
g++ @global.options @absolute.options -MM -MT .csconfig/daefdcfbcd2ec9c_main.d -MT .csconfig/daefdcfbcd2ec9c_main.o -I.csconfig -DCHAMPSIM_BUILD=0xdaefdcfbcd2ec9c -MF .csconfig/daefdcfbcd2ec9c_main.o src/main.cc
g++ @global.options @absolute.options -MM -MT .csconfig/generated_environment.d -MT .csconfig/generated_environment.o -I.csconfig -MF .csconfig/generated_environment.d src/generated_environment.cc
g++ @global.options @absolute.options -MM -MT .csconfig/TEST_main.d -MT .csconfig/TEST_main.o -I.csconfig -DCHAMPSIM_BUILD=0xTEST -MF .csconfig/TEST_main.d src/main.cc
g++ @global.options @absolute.options -I.csconfig -c -o .csconfig/generated_environment.o src/generated_environment.cc
g++ @global.options @absolute.options -I.csconfig -DCHAMPSIM_BUILD=0xdaefdcfbcd2ec9c -c -o .csconfig/daefdcfbcd2ec9c_main.o src/main.cc
g++ -L/champsim/vcpkg_installed/x64-linux/lib -L/champsim/vcpkg_installed/x64-linux/lib/manual-link -o bin/champsim-lab1 .csconfig/address.o .csconfig/bandwidth.o .csconfig/cache.o .csconfig/cache_stats.o .csconfig/champsim.o .csconfig/channel.o .csconfig/chrono.o .csconfig/core_stats.o .csconfig/dram_controller.o .csconfig/dram_stats.o .csconfig/extent.o .csconfig/generated_environment.o .csconfig/json_printer.o .csconfig/daefdcfbcd2ec9c_main.o .csconfig/modules.o .csconfig/ooo_cpu.o .csconfig/operable.o .csconfig/plain_printer.o .csconfig/ptw.o .csconfig/ptw_builder.o .csconfig/register_allocator.o .csconfig/tracereader.o .csconfig/vmem.o .csconfig/modules/branch/bimodal/bimodal.o .csconfig/modules/branch/gshare/gshare.o .csconfig/modules/branch/hashtb_perceptron/hashtb_perceptron.o .csconfig/modules/branch/local_bp/local_bp.o .csconfig/modules/branch/my_perceptron/my_perceptron.o .csconfig/modules/branch/tournament/tournament.o .csconfig/modules/btb/basic_btb/basic_btb.o .csconfig/modules/btb/basic_btb/direct_predictor.o .csconfig/modules/btb/basic_btb/indirect_predictor.o .csconfig/modules/btb/basic_btb/return_stack.o .csconfig/modules/prefetcher/ip_stride/ip_stride.o .csconfig/modules/prefetcher/my_prefetcher/my_prefetcher.o .csconfig/modules/prefetcher/next_line/next_line.o .csconfig/modules/prefetcher/no/no.o .csconfig/modules/prefetcher/spp_dev/spp_dev.o .csconfig/modules/prefetcher/va_ampm_lite/va_ampm_lite.o .csconfig/modules/replacement/drrip/drrip.o .csconfig/modules/replacement/lru/lru.o .csconfig/modules/replacement/random/random.o .csconfig/modules/replacement/ship/ship.o .csconfig/modules/replacement/srrip/srrip.o -lClib11 -lzm -lz -lbz2 -lfmt
root@aaefef21d69a:/champsim#
```

2. A screenshot of the output showing that your ChampSim executable can successfully run an instruction trace and generate simulation statistics.

```
root@aaefef21d69a:/champsim# ./bin/champsim --warmup-instructions 1000000 --simulation-instructions 10000000 traces/binary_search.champsimtrace.xz | tee results-part1.txt
[VMEM] WARNING: physical memory size is smaller than virtual memory size.

*** ChampSim Multicore Out-of-Order Simulator ***
Warmup Instructions: 1000000
Simulation Instructions: 10000000
Number of CPUs: 1
Page size: 4096

Off-chip DRAM Size: 16 GiB Channels: 1 Width: 64-bit Data Rate: 3205 MT/s
Warmup finished CPU 0 instructions: 1000001 cycles: 507661 cumulative IPC: 1.97 (Simulation time: 00 hr 00 min 13 sec)
Warmup complete CPU 0 instructions: 1000001 cycles: 507661 cumulative IPC: 1.97 (Simulation time: 00 hr 00 min 13 sec)
Heartbeat CPU 0 instructions: 10000003 cycles: 17141731 heartbeat IPC: 0.5834 cumulative IPC: 0.5411 (Simulation time: 00 hr 03 min 43 sec)
Simulation finished CPU 0 instructions: 10000001 cycles: 18417731 cumulative IPC: 0.543 (Simulation time: 00 hr 04 min 02 sec)
Simulation complete CPU 0 instructions: 10000001 cycles: 18417731 cumulative IPC: 0.543 (Simulation time: 00 hr 04 min 02 sec)

ChampSim completed all CPUs

=== Simulation ===
CPU 0 runs traces/binary_search.champsimtrace.xz

Region of Interest Statistics

CPU 0 cumulative IPC: 0.543 instructions: 10000001 cycles: 18417731
CPU 0 Branch Prediction Accuracy: 85.77% MPKI: 15.69 Average ROB Occupancy at Mispredict: 36.61
Branch type MPKI
BRANCH_DIRECT_JUMP: 0
BRANCH_INDIRECT: 0
BRANCH_CONDITIONAL: 15.69
BRANCH_DIRECT_CALL: 0
BRANCH_INDIRECT_CALL: 0
BRANCH_RETURN: 0

cpu0->llc TOTAL ACCESS: 99903 HIT: 35762 MISS: 64141 MISS_MERGE: 0
cpu0->llc LOAD ACCESS: 97160 HIT: 33116 MISS: 64044 MISS_MERGE: 0
cpu0->llc RFO ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->llc PREFETCH ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->llc WRITE ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->llc TRANSLATION ACCESS: 2743 HIT: 2646 MISS: 97 MISS_MERGE: 0
cpu0->llc PREFETCH REQUESTED: 0 ISSUED: 0 USEFUL: 0 USELESS: 0
cpu0->llc AVERAGE MISS LATENCY: 175 cycles
cpu0->cpu0_DTLB TOTAL ACCESS: 3068656 HIT: 2841745 MISS: 226911 MISS_MERGE: 113739
cpu0->cpu0_DTLB LOAD ACCESS: 3068656 HIT: 2841745 MISS: 226911 MISS_MERGE: 113739
cpu0->cpu0_DTLB RFO ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB PREFETCH ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB WRITE ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB TRANSLATION ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB PREFETCH REQUESTED: 0 ISSUED: 0 USEFUL: 0 USELESS: 0
```

3. Use the provided basic configuration to build ChampSim successfully.

...

```
root@aaefef21d69a:/champsim# ./config.sh
No configuration specified. Building default ChampSim with no prefetching.
root@aaefef21d69a:/champsim# make -j $(nproc)
g++ @global.options @absolute.options -MM
-MT .csconfig/f45bfff292054888f_main.d
-MT .csconfig/f45bfff292054888f_main.o -I.csconfig
-DCHAMPSIM_BUILD=0xf45bfff292054888f -MF .csconfig/f45bfff292054888f_main.d
src/main.cc
g++ @global.options @absolute.options -MM
-MT .csconfig/generated_environment.d
```

```

-MT .csconfig/generated_environment.o -I.csconfig
-MF .csconfig/generated_environment.d src/generated_environment.cc
g++ @global.options @absolute.options -MM -MT .csconfig/TEST_main.d
-MT .csconfig/TEST_main.o -I.csconfig -DCHAMPSIM_BUILD=0xTEST
-MF .csconfig/TEST_main.d src/main.cc
g++ @global.options @absolute.options -I.csconfig -c
-o .csconfig/generated_environment.o src/generated_environment.cc
g++ @global.options @absolute.options -I.csconfig
-DCHAMPSIM_BUILD=0xf45bff292054888f -c
-o .csconfig/f45bff292054888f_main.o src/main.cc
g++ -L/champsim/vcpkg_installed/x64-linux/lib
-L/champsim/vcpkg_installed/x64-linux/lib/manual-link -o
bin/champsim .csconfig/address.o .csconfig/bandwidth.o .csconfig/cache.o
.csconfig/cache_stats.o .csconfig/champsim.o .csconfig/channel.o .csconf
ig/chrono.o .csconfig/core_stats.o .csconfig/dram_controller.o .csconfig/
dram_stats.o .csconfig/extent.o .csconfig/generated_environment.o .csconf
ig/json_printer.o .csconfig/f45bff292054888f_main.o .csconfig/modules.o
.csconfig/ooo_cpu.o .csconfig/operable.o .csconfig/plain_printer.o .csco
nfig/ptw.o .csconfig/ptw_builder.o .csconfig/register_allocator.o .csconf
ig/tracereader.o .csconfig/vmem.o .csconfig/modules/branch/bimodal/bimoda
l.o .csconfig/modules/branch/gshare/gshare.o .csconfig/modules/branch/has
hed_perceptron/hashed_perceptron.o .csconfig/modules/branch/local_bp/loc
al_bp.o .csconfig/modules/branch/my_perceptron/my_perceptron.o .csconfig/
modules/branch/tournament/tournament.o .csconfig/modules/btb/basic_btb/b
asic_btb.o .csconfig/modules/btb/basic_btb/direct_predictor.o .csconfig/m
odules/btb/basic_btb/indirect_predictor.o .csconfig/modules/btb/basic_bt
b/return_stack.o .csconfig/modules/prefetcher/ip_stride/ip_stride.o .csco
nfig/modules/prefetcher/my_prefetcher/my_prefetcher.o .csconfig/modules/
prefetcher/next_line/next_line.o .csconfig/modules/prefetcher/no/no.o .cs
config/modules/prefetcher/spp_dev/spp_dev.o .csconfig/modules/prefetcher
/va_ampm_lite/va_ampm_lite.o .csconfig/modules/replacement/drrip/drrip.o
.csconfig/modules/replacement/lru/lru.o .csconfig/modules/replacement/r
andom/random.o .csconfig/modules/replacement/ship/ship.o .csconfig/module
s/replacement/srrip/srrip.o -lCLI11 -llzma -lz -lbz2 -lfmt
```

```

4. Use `binary_search` instruction traces to run the generated ChampSim executable. You may first use a small number of warmup and simulation instructions to verify that the simulator works correctly.

```

```small
root@aaefef21d69a:/champsim# ./bin/champsim --warmup-instructions 1000000
--simulation-instructions 10000000 traces/binary_search.champsimtrace.xz
| tee results-part1.txt
```

```

[VMEM] WARNING: physical memory size is smaller than virtual memory size.

\*\*\* ChampSim Multicore Out-of-Order Simulator \*\*\*

Warmup Instructions: 1000000

Simulation Instructions: 10000000

Number of CPUs: 1

Page size: 4096

Off-chip DRAM Size: 16 GiB Channels: 1 Width: 64-bit Data Rate: 3205 MT/s

Warmup finished CPU 0 instructions: 1000001 cycles: 507661 cumulative IPC: 1.97 (Simulation time: 00 hr 00 min 13 sec)

Warmup complete CPU 0 instructions: 1000001 cycles: 507661 cumulative IPC: 1.97 (Simulation time: 00 hr 00 min 13 sec)

Heartbeat CPU 0 instructions: 10000003 cycles: 17141781 heartbeat IPC: 0.5834 cumulative IPC: 0.5411 (Simulation time: 00 hr 03 min 43 sec)

Simulation finished CPU 0 instructions: 10000001 cycles: 18417731 cumulative IPC: 0.543 (Simulation time: 00 hr 04 min 02 sec)

Simulation complete CPU 0 instructions: 10000001 cycles: 18417731 cumulative IPC: 0.543 (Simulation time: 00 hr 04 min 02 sec)

ChampSim completed all CPUs

=== Simulation ===

CPU 0 runs traces/binary\_search.champsimtrace.xz

#### Region of Interest Statistics

CPU 0 cumulative IPC: 0.543 instructions: 10000001 cycles: 18417731

CPU 0 Branch Prediction Accuracy: 85.77% MPKI: 15.69 Average ROB Occupancy at Mispredict: 36.61

Branch type MPKI

BRANCH\_DIRECT\_JUMP: 0

BRANCH\_INDIRECT: 0

BRANCH\_CONDITIONAL: 15.69

BRANCH\_DIRECT\_CALL: 0

BRANCH\_INDIRECT\_CALL: 0

BRANCH\_RETURN: 0

|                 |         |            |             |       |
|-----------------|---------|------------|-------------|-------|
| cpu0->LLC TOTAL | ACCESS: | 99903 HIT: | 35762 MISS: | 64141 |
| MISS_MERGE:     | 0       |            |             |       |
| cpu0->LLC LOAD  | ACCESS: | 97160 HIT: | 33116 MISS: | 64044 |
| MISS_MERGE:     | 0       |            |             |       |
| cpu0->LLC RFO   | ACCESS: | 0 HIT:     | 0 MISS:     | 0     |
| MISS_MERGE:     | 0       |            |             |       |

```

cpu0->LLC PREFETCH ACCESS: 0 HIT: 0 MISS: 0
MISS_MERGE: 0
cpu0->LLC WRITE ACCESS: 0 HIT: 0 MISS: 0
MISS_MERGE: 0
cpu0->LLC TRANSLATION ACCESS: 2743 HIT: 2646 MISS: 97
MISS_MERGE: 0
cpu0->LLC PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->LLC AVERAGE MISS LATENCY: 175 cycles
cpu0->cpu0_DTLB TOTAL ACCESS: 3068656 HIT: 2841745
MISS: 226911 MISS_MERGE: 113739
cpu0->cpu0_DTLB LOAD ACCESS: 3068656 HIT: 2841745
MISS: 226911 MISS_MERGE: 113739
cpu0->cpu0_DTLB RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB TRANSLATION ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_DTLB AVERAGE MISS LATENCY: 7.989 cycles
cpu0->cpu0_ITLB TOTAL ACCESS: 972 HIT: 972
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB LOAD ACCESS: 972 HIT: 972
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB TRANSLATION ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_ITLB AVERAGE MISS LATENCY: - cycles
cpu0->cpu0_L1D TOTAL ACCESS: 3083062 HIT: 2797452
MISS: 285610 MISS_MERGE: 143943
cpu0->cpu0_L1D LOAD ACCESS: 2367522 HIT: 2087482
MISS: 280040 MISS_MERGE: 143482

```

```

cpu0->cpu0_L1D RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1D PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1D WRITE ACCESS: 701134 HIT: 701134
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1D TRANSLATION ACCESS: 14406 HIT: 8836
MISS: 5570 MISS_MERGE: 461
cpu0->cpu0_L1D PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_L1D AVERAGE MISS LATENCY: 97.58 cycles
cpu0->cpu0_L1I TOTAL ACCESS: 972 HIT: 972
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I LOAD ACCESS: 972 HIT: 972
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I TRANSLATION ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_L1I AVERAGE MISS LATENCY: - cycles
cpu0->cpu0_L2C TOTAL ACCESS: 141667 HIT: 41764
MISS: 99903 MISS_MERGE: 0
cpu0->cpu0_L2C LOAD ACCESS: 136558 HIT: 39398
MISS: 97160 MISS_MERGE: 0
cpu0->cpu0_L2C RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L2C PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L2C WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L2C TRANSLATION ACCESS: 5109 HIT: 2366
MISS: 2743 MISS_MERGE: 0
cpu0->cpu0_L2C PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_L2C AVERAGE MISS LATENCY: 126 cycles
cpu0->cpu0_STLB TOTAL ACCESS: 113172 HIT: 105969
MISS: 7203 MISS_MERGE: 0

```

```

cpu0->cpu0_STLB LOAD ACCESS: 113172 HIT: 105969
MISS: 7203 MISS_MERGE: 0
cpu0->cpu0_STLB RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB TRANSLATION ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_STLB AVERAGE MISS LATENCY: 61.68 cycles

```

#### DRAM Statistics

```

Channel 0 RQ ROW_BUFFER_HIT: 50
 ROW_BUFFER_MISS: 64091
 AVG DBUS CONGESTED CYCLE: 2.468
Channel 0 WQ ROW_BUFFER_HIT: 0
 ROW_BUFFER_MISS: 0
 FULL: 0
Channel 0 REFRESHES ISSUED: 1535
...

```

...

[VMEM] WARNING: physical memory size is smaller than virtual memory size.

\*\*\* ChampSim Multicore Out-of-Order Simulator \*\*\*

Warmup Instructions: 100000000

Simulation Instructions: 500000000

Number of CPUs: 1

Page size: 4096

Off-chip DRAM Size: 16 GiB Channels: 1 Width: 64-bit Data Rate: 3205 MT/s  
 Heartbeat CPU 0 instructions: 10000002 cycles: 4084436 heartbeat IPC: 2.448  
 cumulative IPC: 2.448 (Simulation time: 00 hr 01 min 35 sec)  
 Heartbeat CPU 0 instructions: 20000002 cycles: 7940788 heartbeat IPC: 2.593  
 cumulative IPC: 2.519 (Simulation time: 00 hr 03 min 05 sec)  
 Heartbeat CPU 0 instructions: 30000003 cycles: 11798237 heartbeat IPC: 2.592  
 cumulative IPC: 2.543 (Simulation time: 00 hr 04 min 37 sec)  
 Heartbeat CPU 0 instructions: 40000003 cycles: 15656243 heartbeat IPC: 2.592  
 cumulative IPC: 2.555 (Simulation time: 00 hr 06 min 10 sec)

Heartbeat CPU 0 instructions: 50000003 cycles: 19513157 heartbeat IPC: 2.593  
cumulative IPC: 2.562 (Simulation time: 00 hr 07 min 51 sec)  
Heartbeat CPU 0 instructions: 60000005 cycles: 23370057 heartbeat IPC: 2.593  
cumulative IPC: 2.567 (Simulation time: 00 hr 09 min 37 sec)  
Heartbeat CPU 0 instructions: 70000005 cycles: 27227684 heartbeat IPC: 2.592  
cumulative IPC: 2.571 (Simulation time: 00 hr 11 min 22 sec)  
Heartbeat CPU 0 instructions: 80000009 cycles: 31084553 heartbeat IPC: 2.593  
cumulative IPC: 2.574 (Simulation time: 00 hr 13 min 15 sec)  
Heartbeat CPU 0 instructions: 90000013 cycles: 34943254 heartbeat IPC: 2.592  
cumulative IPC: 2.576 (Simulation time: 00 hr 15 min 25 sec)  
Warmup finished CPU 0 instructions: 100000003 cycles: 38799674 cumulative  
IPC: 2.577 (Simulation time: 00 hr 17 min 15 sec)  
Warmup complete CPU 0 instructions: 100000003 cycles: 38799674 cumulative  
IPC: 2.577 (Simulation time: 00 hr 17 min 15 sec)  
Heartbeat CPU 0 instructions: 100000013 cycles: 38799676 heartbeat IPC:  
2.593 cumulative IPC: 5 (Simulation time: 00 hr 17 min 15 sec)  
Heartbeat CPU 0 instructions: 110000014 cycles: 56739482 heartbeat IPC:  
0.5574 cumulative IPC: 0.5574 (Simulation time: 00 hr 21 min 00 sec)  
Heartbeat CPU 0 instructions: 120000016 cycles: 74685197 heartbeat IPC:  
0.5572 cumulative IPC: 0.5573 (Simulation time: 00 hr 25 min 28 sec)  
Heartbeat CPU 0 instructions: 130000016 cycles: 92616294 heartbeat IPC:  
0.5577 cumulative IPC: 0.5574 (Simulation time: 00 hr 29 min 16 sec)  
Heartbeat CPU 0 instructions: 140000019 cycles: 110557992 heartbeat IPC:  
0.5574 cumulative IPC: 0.5574 (Simulation time: 00 hr 32 min 53 sec)  
Heartbeat CPU 0 instructions: 150000021 cycles: 128543286 heartbeat IPC:  
0.556 cumulative IPC: 0.5571 (Simulation time: 00 hr 36 min 01 sec)  
Heartbeat CPU 0 instructions: 160000021 cycles: 146491865 heartbeat IPC:  
0.5571 cumulative IPC: 0.5571 (Simulation time: 00 hr 39 min 24 sec)  
Heartbeat CPU 0 instructions: 170000021 cycles: 164480197 heartbeat IPC:  
0.5559 cumulative IPC: 0.557 (Simulation time: 00 hr 42 min 32 sec)  
Heartbeat CPU 0 instructions: 180000021 cycles: 182375032 heartbeat IPC:  
0.5588 cumulative IPC: 0.5572 (Simulation time: 00 hr 45 min 47 sec)  
Heartbeat CPU 0 instructions: 190000022 cycles: 200319799 heartbeat IPC:  
0.5573 cumulative IPC: 0.5572 (Simulation time: 00 hr 48 min 57 sec)  
Heartbeat CPU 0 instructions: 200000023 cycles: 218338486 heartbeat IPC:  
0.555 cumulative IPC: 0.557 (Simulation time: 00 hr 52 min 07 sec)  
Heartbeat CPU 0 instructions: 210000026 cycles: 236258194 heartbeat IPC:  
0.558 cumulative IPC: 0.5571 (Simulation time: 00 hr 55 min 20 sec)  
Heartbeat CPU 0 instructions: 220000028 cycles: 254180844 heartbeat IPC:  
0.558 cumulative IPC: 0.5572 (Simulation time: 00 hr 58 min 50 sec)  
Heartbeat CPU 0 instructions: 230000030 cycles: 272171446 heartbeat IPC:  
0.5558 cumulative IPC: 0.5571 (Simulation time: 01 hr 01 min 58 sec)  
Heartbeat CPU 0 instructions: 240000031 cycles: 290143332 heartbeat IPC:  
0.5564 cumulative IPC: 0.557 (Simulation time: 01 hr 05 min 06 sec)

\*\*\* Reached end of trace: (0, "traces/binary\_search.champsimtrace.xz")

| Instructions | Cycles    | Heartbeat IPC | Cumulative IPC | Simulation time     |
|--------------|-----------|---------------|----------------|---------------------|
| 25000031     | 308073797 | 0.5577        | 0.5571         | 01 hr 08 min 36 sec |
| 26000035     | 325988131 | 0.5582        | 0.5571         | 01 hr 12 min 58 sec |
| 27000035     | 343856524 | 0.5596        | 0.5573         | 01 hr 17 min 18 sec |
| 28000035     | 361781109 | 0.5579        | 0.5573         | 01 hr 20 min 36 sec |
| 29000037     | 379749375 | 0.5565        | 0.5573         | 01 hr 23 min 43 sec |
| 30000040     | 397633297 | 0.5592        | 0.5574         | 01 hr 26 min 50 sec |
| 31000040     | 415503875 | 0.5596        | 0.5575         | 01 hr 30 min 03 sec |
| 32000042     | 433543262 | 0.5543        | 0.5573         | 01 hr 33 min 21 sec |
| 33000042     | 451551143 | 0.5553        | 0.5572         | 01 hr 36 min 29 sec |
| 34000043     | 469522455 | 0.5564        | 0.5572         | 01 hr 39 min 36 sec |
| 35000046     | 487486622 | 0.5567        | 0.5572         | 01 hr 43 min 07 sec |
| 36000047     | 505419054 | 0.5576        | 0.5572         | 01 hr 47 min 13 sec |
| 37000048     | 523345612 | 0.5578        | 0.5572         | 01 hr 50 min 31 sec |
| 38000048     | 541260670 | 0.5582        | 0.5573         | 01 hr 53 min 33 sec |
| 39000051     | 559208086 | 0.5572        | 0.5573         | 01 hr 56 min 49 sec |
| 40000051     | 577161017 | 0.557         | 0.5572         | 02 hr 00 min 16 sec |
| 41000051     | 595103715 | 0.5573        | 0.5572         | 02 hr 03 min 40 sec |
| 42000051     | 613067288 | 0.5567        | 0.5572         | 02 hr 08 min 16 sec |
| 43000055     | 630937404 | 0.5596        | 0.5573         | 02 hr 11 min 50 sec |
| 44000056     | 648830829 | 0.5589        | 0.5573         | 02 hr 15 min 07 sec |
| 45000059     | 666824035 | 0.5558        | 0.5573         | 02 hr 18 min 14 sec |

Heartbeat CPU 0 instructions: 460000060 cycles: 684735851 heartbeat IPC: 0.5583 cumulative IPC: 0.5573 (Simulation time: 02 hr 21 min 21 sec)  
Heartbeat CPU 0 instructions: 470000063 cycles: 702649873 heartbeat IPC: 0.5582 cumulative IPC: 0.5574 (Simulation time: 02 hr 24 min 28 sec)  
Heartbeat CPU 0 instructions: 480000063 cycles: 720639575 heartbeat IPC: 0.5559 cumulative IPC: 0.5573 (Simulation time: 02 hr 27 min 13 sec)  
Heartbeat CPU 0 instructions: 490000063 cycles: 738586952 heartbeat IPC: 0.5572 cumulative IPC: 0.5573 (Simulation time: 02 hr 29 min 58 sec)  
\*\*\* Reached end of trace: (0, "traces/binary\_search.champsimtrace.xz")  
Heartbeat CPU 0 instructions: 500000065 cycles: 756482843 heartbeat IPC: 0.5588 cumulative IPC: 0.5573 (Simulation time: 02 hr 32 min 50 sec)  
Heartbeat CPU 0 instructions: 510000066 cycles: 774377458 heartbeat IPC: 0.5588 cumulative IPC: 0.5574 (Simulation time: 02 hr 35 min 34 sec)  
Heartbeat CPU 0 instructions: 520000066 cycles: 792221078 heartbeat IPC: 0.5604 cumulative IPC: 0.5575 (Simulation time: 02 hr 38 min 18 sec)  
Heartbeat CPU 0 instructions: 530000066 cycles: 810134164 heartbeat IPC: 0.5583 cumulative IPC: 0.5575 (Simulation time: 02 hr 41 min 02 sec)  
Heartbeat CPU 0 instructions: 540000067 cycles: 828115165 heartbeat IPC: 0.5561 cumulative IPC: 0.5574 (Simulation time: 02 hr 44 min 14 sec)  
Heartbeat CPU 0 instructions: 550000067 cycles: 846008114 heartbeat IPC: 0.5589 cumulative IPC: 0.5575 (Simulation time: 02 hr 47 min 05 sec)  
Heartbeat CPU 0 instructions: 560000068 cycles: 863866184 heartbeat IPC: 0.56 cumulative IPC: 0.5575 (Simulation time: 02 hr 49 min 50 sec)  
Heartbeat CPU 0 instructions: 570000069 cycles: 881882222 heartbeat IPC: 0.5551 cumulative IPC: 0.5575 (Simulation time: 02 hr 52 min 35 sec)  
Heartbeat CPU 0 instructions: 580000069 cycles: 899883682 heartbeat IPC: 0.5555 cumulative IPC: 0.5574 (Simulation time: 02 hr 55 min 22 sec)  
Heartbeat CPU 0 instructions: 590000070 cycles: 917848274 heartbeat IPC: 0.5567 cumulative IPC: 0.5574 (Simulation time: 02 hr 58 min 14 sec)  
Simulation finished CPU 0 instructions: 500000001 cycles: 897024804 cumulative IPC: 0.5574 (Simulation time: 03 hr 01 min 11 sec)  
Simulation complete CPU 0 instructions: 500000001 cycles: 897024804 cumulative IPC: 0.5574 (Simulation time: 03 hr 01 min 11 sec)

ChampSim completed all CPUs

=== Simulation ===

CPU 0 runs traces/binary\_search.champsimtrace.xz

#### Region of Interest Statistics

CPU 0 cumulative IPC: 0.5574 instructions: 500000001 cycles: 897024804  
CPU 0 Branch Prediction Accuracy: 85.84% MPKI: 15.62 Average ROB Occupancy at Mispredict: 36.77

Branch type MPKI

BRANCH\_DIRECT\_JUMP: 0

BRANCH\_INDIRECT: 0

BRANCH\_CONDITIONAL: 15.62

BRANCH\_DIRECT\_CALL: 0

BRANCH\_INDIRECT\_CALL: 0

BRANCH\_RETURN: 0

|                 |         |         |      |         |       |         |
|-----------------|---------|---------|------|---------|-------|---------|
| cpu0->LLC TOTAL | ACCESS: | 5018495 | HIT: | 1969817 | MISS: | 3048678 |
| MISS_MERGE:     | 0       |         |      |         |       |         |

|                |         |         |      |         |       |         |
|----------------|---------|---------|------|---------|-------|---------|
| cpu0->LLC LOAD | ACCESS: | 4879635 | HIT: | 1836491 | MISS: | 3043144 |
| MISS_MERGE:    | 0       |         |      |         |       |         |

|               |         |   |      |   |       |   |
|---------------|---------|---|------|---|-------|---|
| cpu0->LLC RFO | ACCESS: | 0 | HIT: | 0 | MISS: | 0 |
| MISS_MERGE:   | 0       |   |      |   |       |   |

|                    |         |   |      |   |       |   |
|--------------------|---------|---|------|---|-------|---|
| cpu0->LLC PREFETCH | ACCESS: | 0 | HIT: | 0 | MISS: | 0 |
| MISS_MERGE:        | 0       |   |      |   |       |   |

|                 |         |   |      |   |       |   |
|-----------------|---------|---|------|---|-------|---|
| cpu0->LLC WRITE | ACCESS: | 0 | HIT: | 0 | MISS: | 0 |
| MISS_MERGE:     | 0       |   |      |   |       |   |

|                       |         |        |      |        |       |      |
|-----------------------|---------|--------|------|--------|-------|------|
| cpu0->LLC TRANSLATION | ACCESS: | 138860 | HIT: | 133326 | MISS: | 5534 |
| MISS_MERGE:           | 0       |        |      |        |       |      |

|                               |   |          |   |
|-------------------------------|---|----------|---|
| cpu0->LLC PREFETCH REQUESTED: | 0 | ISSUED:  | 0 |
| USEFUL:                       | 0 | USELESS: | 0 |

cpu0->LLC AVERAGE MISS LATENCY: 174.8 cycles

|                       |          |             |         |           |
|-----------------------|----------|-------------|---------|-----------|
| cpu0->cpu0_DTLB TOTAL | ACCESS:  | 153839227   | HIT:    | 142494691 |
| MISS:                 | 11344536 | MISS_MERGE: | 5680180 |           |

|                      |          |             |         |           |
|----------------------|----------|-------------|---------|-----------|
| cpu0->cpu0_DTLB LOAD | ACCESS:  | 153839227   | HIT:    | 142494691 |
| MISS:                | 11344536 | MISS_MERGE: | 5680180 |           |

|                     |         |             |      |   |
|---------------------|---------|-------------|------|---|
| cpu0->cpu0_DTLB RFO | ACCESS: | 0           | HIT: | 0 |
| MISS:               | 0       | MISS_MERGE: | 0    |   |

|                          |         |             |      |   |
|--------------------------|---------|-------------|------|---|
| cpu0->cpu0_DTLB PREFETCH | ACCESS: | 0           | HIT: | 0 |
| MISS:                    | 0       | MISS_MERGE: | 0    |   |

|                       |         |             |      |   |
|-----------------------|---------|-------------|------|---|
| cpu0->cpu0_DTLB WRITE | ACCESS: | 0           | HIT: | 0 |
| MISS:                 | 0       | MISS_MERGE: | 0    |   |

|                             |         |             |      |   |
|-----------------------------|---------|-------------|------|---|
| cpu0->cpu0_DTLB TRANSLATION | ACCESS: | 0           | HIT: | 0 |
| MISS:                       | 0       | MISS_MERGE: | 0    |   |

|                                     |   |          |   |
|-------------------------------------|---|----------|---|
| cpu0->cpu0_DTLB PREFETCH REQUESTED: | 0 | ISSUED:  | 0 |
| USEFUL:                             | 0 | USELESS: | 0 |

cpu0->cpu0\_DTLB AVERAGE MISS LATENCY: 5.862 cycles

|                       |         |             |      |       |
|-----------------------|---------|-------------|------|-------|
| cpu0->cpu0_ITLB TOTAL | ACCESS: | 47350       | HIT: | 47350 |
| MISS:                 | 0       | MISS_MERGE: | 0    |       |

|                      |         |             |      |       |
|----------------------|---------|-------------|------|-------|
| cpu0->cpu0_ITLB LOAD | ACCESS: | 47350       | HIT: | 47350 |
| MISS:                | 0       | MISS_MERGE: | 0    |       |

|                     |         |             |      |   |
|---------------------|---------|-------------|------|---|
| cpu0->cpu0_ITLB RFO | ACCESS: | 0           | HIT: | 0 |
| MISS:               | 0       | MISS_MERGE: | 0    |   |

```

cpu0->cpu0_ITLB PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB TRANSLATION ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_ITLB PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_ITLB AVERAGE MISS LATENCY: - cycles
cpu0->cpu0_L1D TOTAL ACCESS: 154555499 HIT: 140296832
MISS: 14258667 MISS_MERGE: 7179029
cpu0->cpu0_L1D LOAD ACCESS: 118783653 HIT: 104805976
MISS: 13977677 MISS_MERGE: 7156163
cpu0->cpu0_L1D RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1D PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1D WRITE ACCESS: 35055574 HIT: 35055574
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1D TRANSLATION ACCESS: 716272 HIT: 435282
MISS: 280990 MISS_MERGE: 22866
cpu0->cpu0_L1D PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_L1D AVERAGE MISS LATENCY: 93.65 cycles
cpu0->cpu0_L1I TOTAL ACCESS: 47350 HIT: 47350
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I LOAD ACCESS: 47350 HIT: 47350
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I TRANSLATION ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L1I PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_L1I AVERAGE MISS LATENCY: - cycles
cpu0->cpu0_L2C TOTAL ACCESS: 7079638 HIT: 2061143
MISS: 5018495 MISS_MERGE: 0
cpu0->cpu0_L2C LOAD ACCESS: 6821514 HIT: 1941879
MISS: 4879635 MISS_MERGE: 0

```

```

cpu0->cpu0_L2C RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L2C PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L2C WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_L2C TRANSLATION ACCESS: 258124 HIT: 119264
MISS: 138860 MISS_MERGE: 0
cpu0->cpu0_L2C PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_L2C AVERAGE MISS LATENCY: 119.8 cycles
cpu0->cpu0_STLB TOTAL ACCESS: 5664356 HIT: 5306220
MISS: 358136 MISS_MERGE: 0
cpu0->cpu0_STLB LOAD ACCESS: 5664356 HIT: 5306220
MISS: 358136 MISS_MERGE: 0
cpu0->cpu0_STLB RFO ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB PREFETCH ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB WRITE ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB TRANSLATION ACCESS: 0 HIT: 0
MISS: 0 MISS_MERGE: 0
cpu0->cpu0_STLB PREFETCH REQUESTED: 0 ISSUED: 0
USEFUL: 0 USELESS: 0
cpu0->cpu0_STLB AVERAGE MISS LATENCY: 28.46 cycles

```

#### DRAM Statistics

```

Channel 0 RQ ROW_BUFFER_HIT: 1728
 ROW_BUFFER_MISS: 3046950
 AVG DBUS CONGESTED CYCLE: 2.433
Channel 0 WQ ROW_BUFFER_HIT: 0
 ROW_BUFFER_MISS: 0
 FULL: 0
Channel 0 REFRESHES ISSUED: 74752
...

```

5. A screenshot of the successful execution of the ChampSim executable with a provided trace.

```
root@aeef21d69a:/champsim# ./bin/champsim --warmup-instructions 1000000 --simulation-instructions 1000000000 traces/binary_search.champsimtrace.xz | tee results-part1.txt
[VME] WARNING: physical memory size is smaller than virtual memory size.

*** ChampSim Multicore Out-of-Order Simulator ***
Warmup Instructions: 1000000
Simulation Instructions: 1000000000
Number of CPUs: 1
Page size: 4096

Off-chip DRAM Size: 16 GiB Channels: 1 Width: 64-bit Data Rate: 3205 MT/s
Warmup finished CPU 0 instructions: 1000001 cycles: 507661 cumulative IPC: 1.97 (Simulation time: 00 hr 00 min 13 sec)
Warmup complete CPU 0 instructions: 1000001 cycles: 507661 cumulative IPC: 1.97 (Simulation time: 00 hr 00 min 13 sec)
Heartbeat CPU 0 instructions: 10000003 cycles: 17141781 heartbeat IPC: 0.5834 cumulative IPC: 0.5411 (Simulation time: 00 hr 03 min 43 sec)
Simulation finished CPU 0 instructions: 10000001 cycles: 18417731 cumulative IPC: 0.543 (Simulation time: 00 hr 04 min 02 sec)
Simulation complete CPU 0 instructions: 10000001 cycles: 18417731 cumulative IPC: 0.543 (Simulation time: 00 hr 04 min 02 sec)

ChampSim completed all CPUs

=== Simulation ===
CPU 0 runs traces/binary_search.champsimtrace.xz

Region of Interest Statistics

CPU 0 cumulative IPC: 0.543 instructions: 10000001 cycles: 18417731
CPU 0 Branch Prediction Accuracy: 85.77% MPKI: 15.69 Average ROB Occupancy at Mispredict: 36.61
Branch type MPKI
BRANCH_DIRECT_JUMP: 0
BRANCH_INDIRECT: 0
BRANCH_CONDITIONAL: 15.69
BRANCH_DIRECT_CALL: 0
BRANCH_INDIRECT_CALL: 0
BRANCH_RETURN: 0

cpu0->LLC TOTAL ACCESS: 99903 HIT: 35762 MISS: 64141 MISS_MERGE: 0
cpu0->LLC LOAD ACCESS: 97160 HIT: 33116 MISS: 64044 MISS_MERGE: 0
cpu0->LLC RFO ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->LLC PREFETCH ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->LLC WRITE ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->LLC TRANSLATION ACCESS: 2743 HIT: 2646 MISS: 97 MISS_MERGE: 0
cpu0->LLC PREFETCH REQUESTED: 0 ISSUED: 0 USEFUL: 0 USELESS: 0
cpu0->LLC AVERAGE MISS LATENCY: 175 cycles
cpu0->cpu0_DTLB TOTAL ACCESS: 3068656 HIT: 2841745 MISS: 226911 MISS_MERGE: 113739
cpu0->cpu0_DTLB LOAD ACCESS: 3068656 HIT: 2841745 MISS: 226911 MISS_MERGE: 113739
cpu0->cpu0_DTLB RFO ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB PREFETCH ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB WRITE ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB TRANSLATION ACCESS: 0 HIT: 0 MISS: 0 MISS_MERGE: 0
cpu0->cpu0_DTLB PREFETCH REQUESTED: 0 ISSUED: 0 USEFUL: 0 USELESS: 0
```

6. Briefly describe the complete workflow of ChampSim , including configuration, compilation, trace execution, and statistics collection.

## 一、配置

配置内容包括 CPU、指令缓存与数据缓存、分支预测器、预取器以及 DRAM 系统。CPU 模型决定了指令的获取、解码、调度、执行及退役流程。缓存层次结构包括：L1 指令缓存（L1I）、L1 数据缓存（L1D）、L2 缓存、LLC 以及 DRAM。

## 二、编译

使用 config.sh 配置，再用 make 编译

```
./config.sh lab1_config.json
```

```
make -j $(nproc)
```

## 三、执行 trace

ChampSim 执行 trace。运行时通过参数指定 warmup 指令数和 simulation 指令数：

```
./bin/champsim \ --warmup-instructions 100000000 \ --simulation-instructions 500000000 \ traces/binary_search.champsimtrace.xz
```

Warmup 阶段：让 cache 层次和分支预测器进入有代表性的状态，但不计入最终统计。

Simulation 阶段：收集性能统计。

## 四、收集统计

仿真结束后重定向保存到文件，输出中重要的统计包括：IPC：每周期退休指令数；

Cache 统计信息；分支 MPKI：每千条指令的分支误预测数。